

Salahuddin Khan

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PROFESSIONAL SUMMARY

Robotics and perception engineer with 6+ years of experience developing real-time SLAM, multi-view geometry, and sensor fusion systems for robotics and autonomous vehicles. Experienced in building scalable perception pipelines for 3D reconstruction, localization, and mapping using multi-camera and multimodal sensor stacks. Strong background in combining classical geometric vision methods with modern machine learning models to deploy robust perception systems in real-world environments.

EDUCATION

Cornell University 2024 – 2025

M.Eng. Electrical and Computer Engineering

GPA: 3.53

Coursework: Computer Vision, Applied Machine Learning, Digital Signal Processing, Practical ML Applications

NIIT University 2013 – 2017

B.Tech. Electronics and Communication Engineering

GPA: 8.61/10

Honors: National Runner-Up – University CanSat Challenge

TECHNICAL SKILLS

Languages: C, C++, Python, MATLAB

Perception and Geometry: Visual SLAM, Visual-Inertial Odometry, Multi-View Geometry, Structure-from-Motion, Bundle Adjustment, Feature Tracking, Camera Calibration, Pose Graph Optimization

3D Perception: 3D Reconstruction, Point Clouds, Mapping, Sensor Fusion

Frameworks and Libraries: ROS, OpenCV, PCL, Ceres Solver, g2o, PyTorch

Robotics Platforms: PX4, ArduPilot, Embedded ARM Systems

Tools: Linux, Git, AWS

PROFESSIONAL EXPERIENCE

Embedded Robotics Engineer Jan 2022 – Jul 2024

Cartrack, Hyderabad

- Designed and deployed a real-time localization system using inertial sensor fusion (IMU + GPS) based on an invariant Extended Kalman Filter, operating across 200K+ production vehicles.
- Integrated neural network based noise modeling with classical state estimation pipelines to improve robustness under real-world sensing conditions.
- Developed embedded C calibration pipelines for sensor extrinsics and IMU alignment on ARM platforms, reducing factory calibration time by 40%.
- Collaborated with firmware, ML, and backend teams to deploy perception and telemetry pipelines for large-scale vehicle datasets.

Computer Vision Engineer

Aug 2020 – Dec 2021

Preimage, Bangalore

- Developed large-scale 3D reconstruction pipelines using multi-view geometry and bundle adjustment for drone-based mapping.
- Implemented camera calibration, feature matching, and pose graph optimization to recover accurate camera trajectories across large image datasets.
- Built Structure-from-Motion workflows to generate survey-grade 3D terrain models from aerial imagery.

Flight Controls Software Engineer

Mar 2019 – Jul 2020

Third Space Auto, Helsinki

- Developed vision-based navigation and precision landing algorithms for UAVs operating in GPS-denied environments.
- Implemented Visual-Inertial SLAM pipelines integrated into PX4-based autonomous flight stacks.
- Validated perception and navigation systems across 400+ flight sorties in challenging environmental conditions.

Lead Autonomous Systems Engineer

Sep 2017 – Mar 2019

Cell Propulsion, Bangalore

- Led a team developing perception and localization systems for autonomous electric vehicles and indoor mobile robots.
- Designed multi-sensor fusion pipelines combining radar, cameras, IMU, wheel odometry, and GPS for robust vehicle tracking.
- Extended SLAM pipelines to multi-camera configurations and built HD maps for structured road environments.

RESEARCH AND PROJECTS

Real-Time Visual SLAM

Cornell University

- Implemented a monocular visual SLAM pipeline with ORB feature tracking, keyframe management, and bundle adjustment using Python, ROS, and g2o.
- Achieved real-time camera tracking and dense 3D reconstruction on KITTI and EuRoC datasets.

Deep Learning Loop Closure Detection for SLAM

- Benchmarked CNN-based embeddings (ResNet-50, LPIPS, MLP) for loop closure detection in long-horizon SLAM pipelines.
- Achieved 95% accuracy and 0.99 recall on the New College dataset, improving drift correction compared to handcrafted descriptors.

Autonomous Vehicle Perception Stack

- Integrated radar, stereo cameras, and IMU for perception and localization in autonomous electric vehicles.
- Extended multi-camera SLAM pipelines and generated HD vector maps for road environments.

SELECTED AWARDS

National Runner-Up, University CanSat Challenge
3rd Place, Electromania, IIT Kanpur
MIT Global Space Balloon Challenge

INTERESTS

3D Perception, Neural Rendering, Sensor Fusion, Differentiable Geometry, Robotics and Autonomous Systems